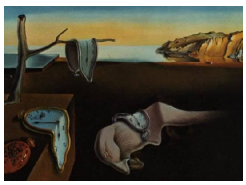

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A Short Demo Article: Regression Models for Count Data in R

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Abstract

MixTwice is an *R* package designed for large-scale differential effects analysis that employs a nonparametric empirical-Bayes framework to jointly model effect sizes and variance heterogeneity. Building on the two-layer mixture structure of MixTwice, we implement new fitting procedures that emphasize computational speed and efficiency. While many empirical-Bayes methods exist in the scientific literature for effects analysis, we illustrate the robustness of this nonparametric grid-based mixture prior approach through simulation.

Keywords: nonparametric empirical Bayes, grid-based mixture prior, unimodal, R.

1. Introduction

Given n_1, n_2 individuals in two groups and m effects of interest, the observed data consist of the between group effect estimates, usually calculated as difference in group means, $\mathbf{x} = (x_1, ..., x_m)$; and the variance estimates of effects, usually calculated as the pooled variance of measurements, $\mathbf{s^2} = (s_1^2, ..., s_m^2)$. The unknown ("true") effects are $\theta = (\theta_1, ..., \theta_m)$, with associated variances $\boldsymbol{\sigma^2} = (\sigma_1^2, ..., \sigma_m^2)$. We aim to perform inference on effects θ_i by borrowing strength across all data pairs (x_i, s_i^2), and to test the null hypothesis $H_{0,i} : \theta_i = 0$.

As traditional parametric models fail to capture complex effect and variance structure, as well as small samples make variance estimates s_i^2 unreliable, MixTwice proposes a two-layer empirical-Bayes framework: (i) lays down finite grids $\{a_{-K}, a_{-K+1}, ..., a_0, a_1, ..., a_K\}$ for θ_i, with $a_0 = 0$ representing the null effect in the testing problem; and $\{0 < b_1, ..., b_L\}$ for σ_i^2 (ii) assumes Gaussian sampling distribution for effect estimates, $x_i|(\theta_i, \sigma_i^2) \overset{\text{ind.}}{\sim} \mathcal{N}(\theta_i, \sigma_i^2)$; and scaled chi-squared sampling distribution for variance estimate, $s_i^2|\sigma_i^2 \overset{\text{ind.}}{\sim} \frac{\sigma_i^2}{\nu}\chi_\nu^2$, with the degrees of freedom $\nu = n_1 + n_2 - 2$ for two group comparison. As a consequence, $x_i \perp\!\!\!\perp s_i^2|(\theta_i, \sigma_i^2)$ (iii) proposes nonparametric mixing distributions `g` and `h` on finite grids, such that

$\mathtt{g}_k = P(\theta_i = a_k), \mathtt{h}_l = P(\sigma_i^2 = b_l)$, and assumes prior independence of the latent effects and variances, i.e. $P(\theta_i = a_k, \sigma_i^2 = b_l) = \mathtt{g}_k \mathtt{h}_l$ (iv) estimates the mixing weights $\mathtt{g}_k$ and $\mathtt{h}_l$ via constrained maximum likelihood. Under these assumptions, $\mathbf{x}$ and $\mathbf{s^2}$ are unbiased estimates of θ and $\boldsymbol{\sigma^2}$.

Constraints are crucial to MixTwice, comprising fundamental probability constraints with a special focus on unimodal shape restriction on effect sizes prior $\mathtt{g}$ that stabilize estimations and improve interpretability while preserving flexibility:

- To form valid probability distributions for effect sizes and underlying variances:

$$\begin{aligned} &\mathtt{g}_k \geq 0, \quad \mathtt{h}_l \geq 0 \qquad \forall k \in \{-K, ..., 0, ..., K\}, \quad l \in \{1, ..., L\} \\ &\sum_k \mathtt{g}_k = 1, \quad \sum_l \mathtt{h}_l = 1 \end{aligned} \tag{1}$$

- To reflect sparsity belief on true effects, enforce an effect-size prior $\mathtt{g}$ that is unimodal at the null hypothesis with monotonically non-increasing tails:

$$\begin{aligned} &\mathtt{g}_k \leq \mathtt{g}_{k+1}, \quad k \in \{-K, -K+1, ..., -1\} \\ &\mathtt{g}_k \geq \mathtt{g}_{k+1}, \quad k \in \{0, 1, ..., K-1\} \end{aligned} \tag{2}$$

Notably, there are minimal constraints on the variance prior $\mathtt{h}$, allowing a data-driven estimate that captures the true variance structure and enables better precision-adaptive shrinkage of effect sizes. Eventually, in this multiple-testing setting, MixTwice does not rely on p-values. Instead, it fits an empirical Bayes mixture model across all peptides and assesses evidence using posterior error rates:

- local false discovery rate (lfdr) - the posterior probability that the effect is null

$$\text{lfdr}_i = P(\theta_i = a_0 \mid x_i, s_i^2)$$

- local false sign rate (lfsr) - the posterior probability of an incorrect sign call

$$\text{lfsr}_i \;=\; \min\Big\{\sum_{k \leq 0} P(\theta_i = a_k \mid x_i, s_i^2),\; \sum_{k \geq 0} P(\theta_i = a_k \mid x_i, s_i^2)\Big\}$$

- posterior mean - shrinkage estimate of the effect size

$$\hat{\theta}_i = E[\theta_i | x_i, s_i^2]$$

- posterior standard deviation - uncertainty in the effect estimate

$$\text{SD}\Big(\theta_i \mid x_i, s_i^2\Big) = \sqrt{\text{Var}(\theta_i \mid x_i, s_i^2)}\,.$$

- 95% posterior credible interval - an interval contains the true effect size with 0.95 posterior probability

$$\mathrm{CI}_{0.95}(\theta_i) = \left[Q_{0.025}\left(\theta_i \mid x_i, s_i^2\right),\ Q_{0.975}\left(\theta_i \mid x_i, s_i^2\right) \right].$$

We updated the R package **MixTwice**, which implements the proposed empirical Bayes mixture model for large-scale effect size estimations, and posterior inference.

The rest of the article is structured as follows. In section 2, we describe the implementation detail of the R function `mixtwice_nloptr()` in **MixTwice**, including grid construction, constrained maximum likelihood estimation, and the underlying optimization procedure. Section 3 evaluates the empirical performance of MixTwice, focusing on shrinkage behavior, posterior error rates, and robustness, comparing it with the two-step ASH approach, which models only the effect-size distribution under a similar unimodal prior.

2. The MixTwice package: implementation and usage

2.1. Grid construction

The effect grid $\{a_{-K}, a_{-K+1}, \ldots, a_0, a_1, \ldots, a_K\}$ is a finite collection of candidate values for underlying true effects θ_i. As effect sizes in high-dimensional settings are typically concentrated near zero, we construct a symmetric grid centered at zero with $2K+1$ points. To provide flexible coverage of plausible effect sizes, we define the boundary $c = 1.1\ \max_i |x_i|$. The grid points are then evenly spaced between $-c$ and c according to $a_j = (\frac{j}{K})\, c$ for $j = -K, \ldots, K$. This construction guarantees that the central grid point is exactly $a_0 = 0$, allowing the model to represent a spike at zero while maintaining sufficient flexibility for estimating the continuous non-null component.

The variance grid $\{b_1, \ldots, b_L\}$ is a finite collection of candidate values for the underlying variances σ_i^2. As variance levels in high-dimensional settings can vary substantially across observations, we construct a grid that spans the range of the observed variance estimates. Specifically, we define the lower and upper bounds as $\sqrt{\min_i s_i^2}$ and $\sqrt{\max_i s_i^2}$, respectively. The grid points are then evenly spaced between these bounds on the standard deviation scale, and squared to obtain variance values: $b_l = \sigma_l^2, \quad \sigma_l \in \left[\sqrt{\min_i s_i^2}, \sqrt{\max_i s_i^2}\right], \quad l = 1, \ldots, L$ This construction provides flexible coverage of plausible variance levels while allowing the model to capture both low- and high-variance components.

2.2. Constrained maximum likelihood

With total m observed effects $\mathbf{x}$ and standard errors $\mathbf{s}$, define finite grids $\{a_{-K}, a_{-K+1}, ..., a_0 = 0, a_1, ..., a_K\}$ for θ_i, $\{0 < b_1, ..., b_L\}$ for σ_i^2. Let $\mathtt{g}$ and $\mathtt{h}$ denote the corresponding mixing distributions that satisfy previously stated assumptions and constraints. The marginal likelihood is derived conditional on $\mathtt{g}$ and $\mathtt{h}$, which are subsequently estimated via constrained maximum likelihood.

$$
\begin{aligned}
p_i = p(x_i, s_i^2 \mid \mathtt{g}, \mathtt{h}) &= \sum_k \sum_l p\Big(x_i, s_i^2, \theta_i = a_k, \sigma_i^2 = b_l \mid \mathtt{g}, \mathtt{h}\Big) \\
&= \sum_k \sum_l p\Big(x_i, s_i^2 \mid \theta_i = a_k, \sigma_i^2 = b_l, \mathtt{g}, \mathtt{h}\Big) \Pr(\theta_i = a_k \mid \mathtt{g}) \Pr(\sigma_i^2 = b_l \mid \mathtt{h}) \\
&= \sum_k \sum_l p\Big(x_i, s_i^2 \mid \theta_i = a_k, \sigma_i^2 = b_l\Big) \mathtt{g}_k \mathtt{h}_l \\
&\quad x_i \text{ and } s_i^2 \text{ are conditionally independent given } (\theta_i, \sigma_i^2): \\
&= \sum_k \sum_l p\Big(x_i \mid \theta_i = a_k, \sigma_i^2 = b_l\Big)\ p\Big(s_i^2 \mid \theta_i = a_k, \sigma_i^2 = b_l\Big)\ \mathtt{g}_k \mathtt{h}_l \\
&\quad \text{By modeling distribution assumption}: \\
&= \sum_k \sum_l \frac{1}{\sqrt{b_l}} \phi\Big(\frac{x_i - a_k}{\sqrt{b_l}}\Big) \mathtt{g}_k \frac{\nu}{b_l} \chi_\nu^2\Big(\frac{\nu s_i^2}{b_l}\Big) \mathtt{h}_l
\end{aligned}
$$

For computational efficiency, we then implement two matrices. For each a_k value in the effect-size grid, build a $p \times L$ matrix whose entry $N_k[i, \ell] = p(x_i \mid \theta_i = a_k, \sigma_i^2 = b_l)$ for every (i, b_l) pair. Similarly, build one $p \times L$ matrix whose entry $D[i, l] = p(s_i^2 \mid \sigma_i^2 = b_l)$ for all b_l values in the variance grid. Then the marginal likelihood can be calculated through matrix manipulations. For each a_k, we calculated the vector A_k where each entry is $A_k(i) = \sum_l h_l D[i, l] N_k[i, l]$. The marginal likelihood is updated sequentially by summing $A_k(i)\mathtt{g}_k$, so $p_i = \sum_k A_k(i)\mathtt{g}_k$.

2.3. Estimating the mixing distributions

The discrete mixing distributions $\mathtt{g}$ and $\mathtt{h}$ are estimated by constrained maximum likelihood (equivalently by minimizing constrained negative log-likelihood), subject to the probability and shape constraints, in equations 1 and 2.

$$
\min_{\mathtt{g},\mathtt{h}} \{-\ell(\mathtt{g}, \mathtt{h}) \ = \ -\sum_{i=1}^{m} \log\big(p(x_i, s_i^2 \mid \mathtt{g}, \mathtt{h})\big)\} = -\sum_{i=1}^{m} log(p_i)
$$

The analytic gradients of log-likelihood $\ell(\mathtt{g}, \mathtt{h})$ with respect to $\mathtt{g}_k$ and $\mathtt{h}_l$ are derived as follows:

$$
\begin{aligned}
\frac{\partial \ell}{\partial \mathtt{g}_k} &= \sum_{i=1}^{m} \frac{A_k(i)}{p_i} \\
\frac{\partial \ell}{\partial \mathtt{h}_l} &= \sum_{i=1}^{m} \frac{\sum_k \mathtt{g}_k\, D[i, l]\, N_k[i, l]}{p_i}
\end{aligned}
$$

The estimators $\hat{\mathtt{g}}$ and $\hat{\mathtt{h}}$ are obtained by solving a constrained nonlinear optimization problem with the **nloptr** package, which provides an R interface to the NLopt library of numerical solvers. $\hat{\mathtt{g}}$ and $\hat{\mathtt{h}}$ are chosen to minimize the negative log-likelihood of the observed statistics while enforcing normalization, nonnegativity, and unimodality so that both $\mathtt{g}$ and $\mathtt{h}$ represent

valid probability distributions. The optimization problem is cast in the standard nloptr form:

$$\begin{aligned}
\min_{(\mathtt{g},\mathtt{h})} \quad & f(\mathtt{g},\mathtt{h}) = -\ell(\mathtt{g},\mathtt{h}) = -\sum_{i=1}^{m} \log p(x_i, s_i^2 \mid \mathtt{g},\mathtt{h}) \\
\text{s.t.} \quad & \mathtt{g}_k - \mathtt{g}_{k+1} \le 0, \quad k \in \{-K,\dots,-1\}, \\
& -\mathtt{g}_k + \mathtt{g}_{k+1} \le 0, \quad k \in \{0,\dots,K-1\}, \\
& -\mathtt{g}_k \le 0,\ -\mathtt{h}_l \le 0, \qquad \forall k,l, \\
& \sum_k \mathtt{g}_k - 1 = 0, \quad \sum_l \mathtt{h}_l - 1 = 0, \\
& 0 \le \mathtt{g}_k,\ \mathtt{h}_l \le 1.
\end{aligned}$$

2.4. Optimization

For each observation i and grid pair (a_k, b_l), define

$$p_{ikl} = p(x_i \mid \theta_i = a_k, \sigma_i^2 = b_l)\, p(s_i^2 \mid \sigma_i^2 = b_l).$$

Then the marginal likelihood and optimization objective function become:

$$p_i(g,h) = \sum_{k,l} g_k h_l\, p_{ikl}, \quad f(g,h) = -\sum_{i=1}^{m} \log p_i(g,h).$$

The optimization procedure uses a short EM–PAVA warm start (20 iterations) followed by constrained likelihood optimization via SLSQP.

Step 1: Initialize mixing proportions.

Initialize g and h as uniform probability vectors:

$$g_k^{(0)} = \frac{1}{2K+1}, \qquad h_l^{(0)} = \frac{1}{L}.$$

Step 2: EM–PAVA iterations.

For $t = 1,\dots,20$, update:

- *E-step:*

$$w_{ikl}^{(t)} = \frac{g_k^{(t)} h_l^{(t)} p_{ikl}}{\sum_{k',l'} g_{k'}^{(t)} h_{l'}^{(t)} p_{ik'l'}}.$$

- *M-step:*

$$g_k^{(t+1)} = \frac{1}{m}\sum_{i=1}^{m}\sum_{l} w_{ikl}^{(t)}, \qquad h_l^{(t+1)} = \frac{1}{m}\sum_{i=1}^{m}\sum_{k} w_{ikl}^{(t)}.$$

- *PAVA step:*

  Adjust $g^{(t+1)}$ to satisfy the unimodal constraint

$$g_{-K} \le \cdots \le g_0 \ge \cdots \ge g_K,$$

by replacing it with

$$g^{(t+1)} = \arg\min_{g \in \mathcal{U}} \sum_k \left(g_k - g_k^{(t+1)}\right)^2, \quad \mathcal{U} = \{g : g_{-K} \leq \cdots \leq g_0 \geq \cdots \geq g_K\}.$$

Step 3: Constrained likelihood optimization.
After 20 EM–PAVA iterations, we obtain a warm start $(g^{\text{warm}}, h^{\text{warm}})$. Starting from this initialization, we solve

$$\min_{g,h} -\sum_{i=1}^{m} \log\left(\sum_{k,l} g_k h_l p_{ikl}\right)$$

subject to

$$\sum_k g_k = 1, \quad \sum_l h_l = 1, \quad g_k, h_l \geq 0,$$
$$g_{-K} \leq \cdots \leq g_0 \geq \cdots \geq g_K,$$

using the SLSQP algorithm via the `nloptr` package, which performs gradient-based constrained optimization.
The gradients used are

$$\frac{\partial f}{\partial g_k} = -\sum_{i=1}^{m} \frac{\sum_l h_l p_{ikl}}{p_i(g,h)}, \qquad \frac{\partial f}{\partial h_l} = -\sum_{i=1}^{m} \frac{\sum_k g_k p_{ikl}}{p_i(g,h)},$$

leading to the final estimates $(\hat{g}, \hat{h})$.

2.5. Posterior inference

We draw inference on true effect size and variance through the observed data pairs and the estimated mixing distributions. We compute the posterior probability for θ_i and σ_i^2 belonging to one of the grid values, as

$$\pi_{ikl} = p\Big(\theta_i = a_k, \sigma_i^2 = b_l \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}\Big) = \frac{\frac{1}{\sqrt{b_l}}\phi(\frac{x_i - a_k}{\sqrt{b_l}})\hat{g}_k \frac{\nu}{b_l}\chi_\nu^2(\frac{\nu s_i^2}{b_l})\hat{h}_l}{\sum_{k'}\sum_{l'} \frac{1}{\sqrt{b_{l'}}}\phi(\frac{x_i - a_{k'}}{\sqrt{b_{l'}}})\hat{g}_{k'} \frac{\nu}{b_{l'}}\chi_\nu^2(\frac{\nu s_i^2}{b_{l'}})\hat{h}_{l'}}$$

The point estimation of the true effect size θ_i is obtained through the posterior mean:

$$\mathbb{E}[\theta_i \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}] = \sum_k \sum_l a_k\, \pi_{ikl} = x_i - \sum_k \sum_l (x_i - a_k)\pi_{ikl} = \sum_k \sum_l a_k \pi_{ikl}$$

The 95% credible interval of the true effect size θ_i is obtained through integrating the posterior distribution of the effect size $(P(\theta_i \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}) = \sum_l \pi_{ikl})$ over the finite grid:

$$\left[\min\left\{a_k : \sum_{k' \leq k} \sum_l \pi_{ik'l} \geq 0.025\right\},\ \min\left\{a_k : \sum_{k' \leq k} \sum_l \pi_{ik'l} \geq 0.975\right\}\right].$$

The uncertainty in effect size point estimation is addressed in the posterior standard deviation:

$$\mathrm{SD}\Big(\theta_i \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}\Big) = \sqrt{\mathbb{E}[\theta_i^2 \mid x_i, s_i^2, \hat{g}, \hat{h}] - \Big(\mathbb{E}[\theta_i \mid x_i, s_i^2, \hat{g}, \hat{h}]\Big)^2} = \sqrt{\sum_k \sum_l a_k^2 \pi_{ikl} - \left(\sum_k \sum_l a_k \pi_{ikl}\right)^2}$$

To quantify the uncertainty in detecting null effects and the likelihood of incorrect effect size sign calls, we use the following two metrics:

$$\mathrm{lfdr}_i = P(\theta_i = a_0 \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}) = \sum_l \pi_{i0l}$$

$$\mathrm{lfsr}_i = \min\Big\{\sum_{k\le 0} P(\theta_i = a_k \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}),\ \sum_{k\ge 0} P(\theta_i = a_k \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}})\Big\} = \min\left(\sum_{k\le 0}\sum_l \pi_{ikl}, \sum_{k\ge 0}\sum_l \pi_{ikl}\right)$$

2.6. The `mixtwice_nloptr()` function

For the package **MixTwice** (...link) to be loaded into R,

```
R> install.packages("MixTwice")
R> library("MixTwice")
```

The main function in the package, `mixtwice_nloptr()` provides posterior inference for large-scale hypothesis testing on effect sizes between two groups using the information of observed variances through a mixture modeling approach. It has the following input arguments:

- `thetaHat`: The vector of observed effects between two groups. $\mathbf{x} = (x_1, ..., x_m)$.
- `s2`: The vector of observed variance on effect sizes. $\mathbf{s} = (s_1, ..., s_m)$
- `Btheta`: Corresponding to the K in the effect size (θ_i) grid. It defines a symmetric around 0 grid with total 2Btheta+1 values.
- `Bsigma2`: Corresponding to the L in the variance (σ_i^2) grid. It defines a positive grid with total Bsigma2 values.
- `df`: The degrees of freedom of (ν) in the chi-squared distribution of the observed variance conditional on the true unobserved variance, calculated as $n_1 + n_2 - 2$.

The `mixtwice_nloptr()` outputs present the following information:

- Grid values (`grid.theta`, `grid.sigma2`): Finite grid values for effect sizes and variances of effects. See Section 2.1.
- Mixture distributions (`mix.theta`, `mix.sigma2`): Estimated posterior probabilities of grid values through constrained nonlinear optimization, $\hat{\mathtt{g}}_k = P(\theta_i = a_k|(x_i, s_i^2)), \hat{\mathtt{h}}_l = P(\sigma_i^2 = b_l|(x_i, s_i^2))$.

- Posterior probabilities and estimations: See Section 2.4.
  - Posterior point estimations (`pi0`, `post_prob`, `post_mean`, `post_sd`): Corresponding to π_{i0l}, π_{ikl}, $\mathbb{E}[\theta_i \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}}]$, $\mathrm{SD}(\theta_i \mid x_i, s_i^2, \hat{\mathtt{g}}, \hat{\mathtt{h}})$
  - Posterior credible interval (`q025`, `q0975`): Corresponding to the 95% credible intervals for each effect size θ_i.
  - Significance measurements (`lfdr`, `lfsr`): Local false discovery rate and local false sign rate for each effect size θ_i.

3. Comparison with effect-size-only modeling approaches

In this section, we compare the performance of MixTwice, which models both effect size and variance, with existing methods, such as (two-step) ASH, another Bayesian approach that makes similar but additional model assumptions, focusing only on modeling the effect size.

In our notation, we observe m effect sizes, $\mathbf{x} = (x_1, \ldots, x_m)$, for the true effects $\theta = (\theta_1, \ldots, \theta_m)$ from two groups with sample sizes n_1 and n_2. Additionally, we observe the variances of the effect sizes, $\mathbf{s^2} = (s_1^2, \ldots, s_m^2)$, corresponding to the true variances $\sigma^{\mathbf{2}} = (\sigma_1^2, \ldots, \sigma_m^2)$.

3.1. Comparator method: ASH

Assume the unimodal prior distribution on effect sizes θ, with parameters $\pi = (\pi_0, \pi_1, ..., \pi_K)$:

$$g(\theta_i; \pi) = \pi_0 \delta_0(\theta_i) + \sum_{k=1}^{K} \pi_k N(\theta_i; 0, \sigma_k^2).$$

Where π_0 represents the prior probability that the effect θ_i is 0; π_k represents the mixing proportions for each component in the mixture distribution; $\delta_0(\theta_i)$ represents the dirac delta function centered at 0.

The likelihood of $\mathbf{x}$ is modeled with a normal distribution:

$$p(x_i | \theta_i, s_i^2) \sim \mathcal{N}(\theta_i, s_i^2)$$

In R, we implement the method and obtain the posterior inferences, using the **ASH** package:

```
R> ash_result <- ash(thetaHat, sqrt(s2), method = "fdr")
R> ash_result$result$PosteriorMean
R> ash_result$result$PosteriorMean
R> ash_result$result$lfdr
R> ash_result$result$lfsr
```

3.2. Comparator method: two-step ASH

Step 1: Preprocess standard errors through Empirical Bayes shrinkage

Assume a prior inverse-gamma distribution on the true variances of the effects σ_i^2:

$$\sigma_i^{-2} \sim \text{Inverse-Gamma}(\alpha_0, \beta_0)$$

Obtain the adjusted standard errors, $\tilde{s}_i^2$:

$$\tilde{s}_i^2 = \frac{\nu_0 s_0^2 + \nu_i s_i^2}{\nu_0 + \nu_i}$$

Where:

- ν_0: the prior degrees of freedom, reflecting prior knowledge about θ_i.
- s_0^2: the prior variance, reflecting the prior belief about the variability in θ_i.
- ν_i: the degrees of freedom associated with each effect, calculated as $n_1 + n_2 - 2$ in two-sample comparison.
- s_i^2: the observed variance of effect sizes.

In R, we implement step 1 using the function `squeezeVar()` from the **limma** package. The ν_0 and s_0^2 will be automatically estimated through empirical Bayes methods. The observed variance are stored in the vector `s2`, we can then obtain the adjusted standard errors:

```
R> limma::squeezeVar(var = s2, df = n1 + n2 - 2)$var.post
```

Step 2: Empirical Bayes T-Means Model

Assume the unimodal prior distribution on effect sizes θ, with parameters $\pi = (\pi_0, \pi_1, ..., \pi_K)$:

$$g(\theta_i; \pi) = \pi_0 \delta_0(\theta_i) + \sum_{k=1}^{K} \pi_k N(\theta_i; 0, \sigma_k^2).$$

Where π_0 represents the prior probability that the effect θ_i is 0; π_k represents the mixing proportions for each component in the mixture distribution; $\delta_0(\theta_i)$ represents the dirac delta function centered at 0.

The likelihood of $\mathbf{x}$ is modeled with a scaled t-distribution to account for uncertainty in the estimated variances:

$$p(x_i|\theta_i, s_i^2) \sim t_{\nu_i^{\text{post}}} \left(\text{mean} = \theta_i, \text{ variance} = s_i^2\right)$$

Where $\nu_i^{\text{post}} = \nu_0 + \nu_i$ as a result from step 1:

```
R> df_post <- limma::squeezeVar(var = s2, df = n1 + n2 - 2)$df.prior + (n1 + n2 - 2)
```

In R, we implement step 2 using the function `ash()` from the **ashr** package. All posterior information can be obtained:

```
R> ash_result <- ash(thetaHat, se_post, lik = lik_t(df = df_post), method = "fdr")
R> ash_result$result$PosteriorMean
R> ash_result$result$PosteriorMean
R> ash_result$result$lfdr
R> ash_result$result$lfsr
```